CS-860: ARTIFICIAL INTELLIGENCE

- **Introduction**
  - Course Outline
  - AI Approaches
  - AI Implications
  - AI Hype

Text : Artificial Intelligence a Modern Approach
by Stuart Russel and Peter Norvig

INSTRUCTOR:
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Introduction

- **Course Format**
  - **Outline**

Course Outline

Course Information	
Course Number and Title:	CS 860 Artificial Intelligence
Credits:	3 (3+0)
Instructor:	Professor Dr. Kunwar Faraz Ahmed Khan
Course type:	Lecture
Core / Elective:	Core
Course pre-requisites	None
Degree and Semester	PG Course
Month and Year	Spring 2026

Course Outline

Course Assessment		
Exams:	1 Midterm and 1 Final	
Homework:	2-3 Assignments	
Design Projects:	2 Projects	
Quizzes:	6 Quizzes	
Grading(Tentative):	Quizzes:	10 %
	Assignments:	10 %
	Midterm:	25 %
	Final Exam:	40%
	Projects:	15 %

- Course Format
  - Outline

Course Outline

Course book and Related Course Material	
Textbooks:	Artificial Intelligence: A Modern Approach, by Stuart Russell and Peter Norvig, 4th Edition, Pearson, 2020.
Reference Books:	1. Artificial Intelligence: A Systems Approach, by M. Tim Jones, Infinity Science Press, 2008. 2. Pattern Classification, by Richard O. Duda, Peter E. Hart and David G. Stork, Wiley Inter-science, 2nd edition, 2006. 3. Pattern Recognition and Machine Learning, by Christopher Bishop, Springer, 2006.

- Course Format
  - Outline

Course Outline

Course Objectives

- Understand the foundations and historical evolution of Artificial Intelligence as a scientific discipline.
- Formulate real-world problems as AI problems using states, actions, goals, and performance measures.
- Design and analyze rational agents operating in a variety of task environments.
- Apply classical AI techniques such as search and decision-making to solve complex problems.
- Develop critical insight into the limitations, ethical implications, and societal impact of AI systems.
- Build a conceptual foundation for advanced AI topics and research.

Introduction

- **Course Format**
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Course Outline

Ser	Topic	Week
1.	Introduction to Artificial Intelligence: History, Scope, and Definitions	1
2.	Rational Agents and Performance Measures	2
3.	Properties of Task Environments /Types of Intelligent Agents	3
4.	Problem Formulation and State Space Representation	4
5.	Uninformed Search Algorithms (BFS, DFS, UCS)	5-6
6.	Informed Search and Heuristics	7-8
7.	Midterm Exam	9
8.	Shortest Path Algorithms	10-11
9.	Planning Under Uncertainty	12-13
10.	Multi-Agent Systems & Game-Theoretic AI	14-15
11.	Reinforcement Learning	16-17
12.	Final Exam	18

Introduction to AI

- What is Artificial Intelligence?
- A Brief History
- Ethical Aspects of AI
- Biases

Introduction

Brief History

- 1940-1950: Early days
  - 1943: McCulloch & Pitts: Boolean circuit model of brain
  - 1950: Turing's "Computing Machinery and Intelligence"
- 1950—70: Excitement: Look, Ma, no hands!
  - 1950s: Early AI programs, including Samuel's checkers program, Newell & Simon's Logic Theorist, Gelernter's Geometry Engine
  - 1956: Dartmouth meeting: "Artificial Intelligence" adopted
  - 1965: Robinson's complete algorithm for logical reasoning
- 1970—90: Knowledge-based approaches
  - 1969—79: Early development of knowledge-based systems
  - 1980—88: Expert systems industry booms
  - 1988—93: Expert systems industry busts: "AI Winter"
- 1990—: Statistical approaches
  - Resurgence of probability, focus on uncertainty
  - General increase in technical depth
  - Agents and learning systems… "AI Spring"?
- 2010-: Neural networks (deep learning)
  - Great progress in vision, NLP.. "AI Summer"?

Big Scientific Questions of Our Time

- How did the universe originate?
- How did life on Earth originate?
- What is the nature of Intelligence?

The Nature of Intelligence

- *How can we Study it?*
  - Study human (or animal) behavior — Psychology
  - Study human (or animal) brains — Neuroscience
  - Think about it — Philosophy
  - Build and analyze intelligent artifacts — Computer Science

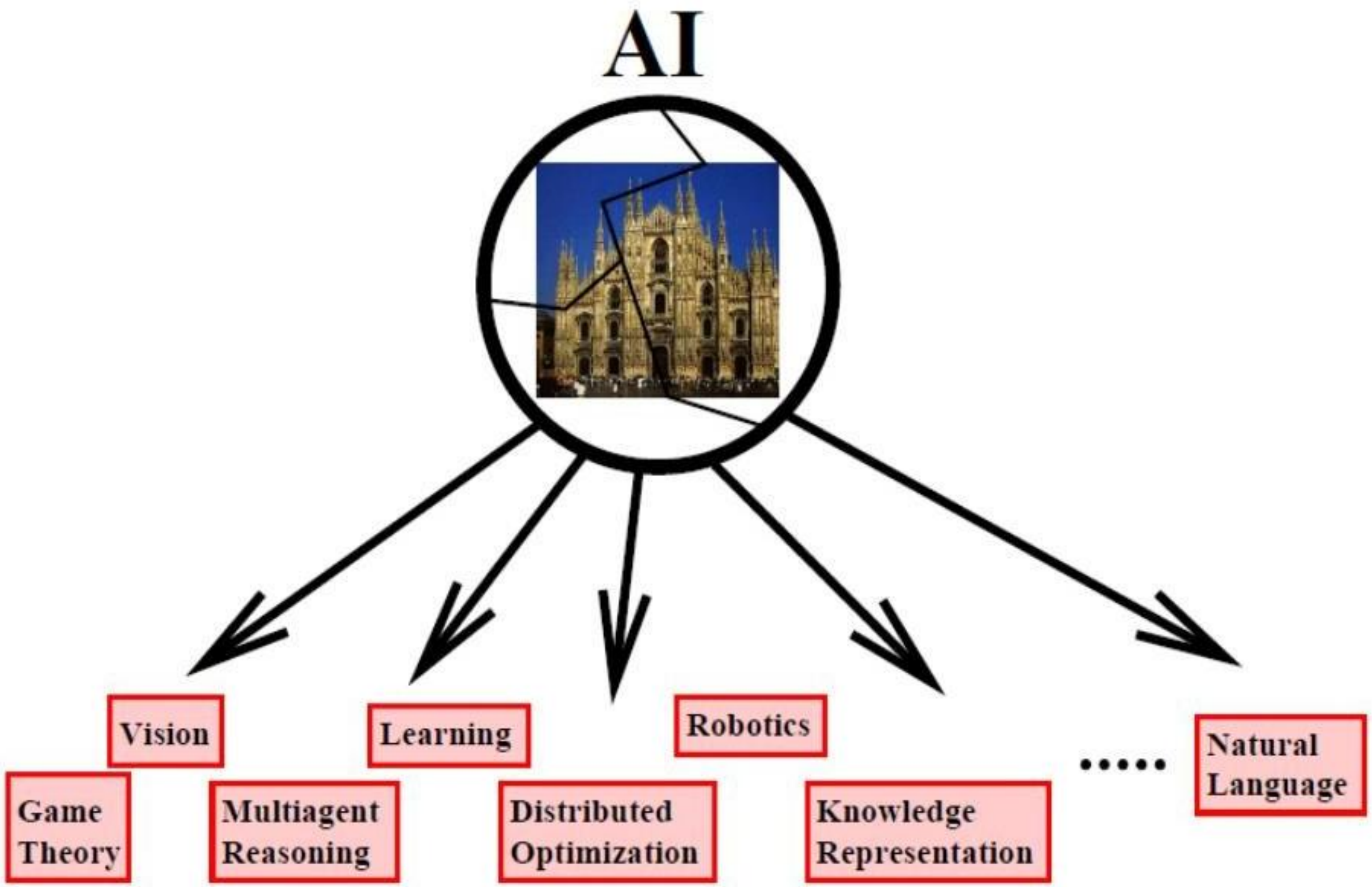
Building Intelligent Artifacts

- *A goal of AI: Robust, fully autonomous agents in the real world*

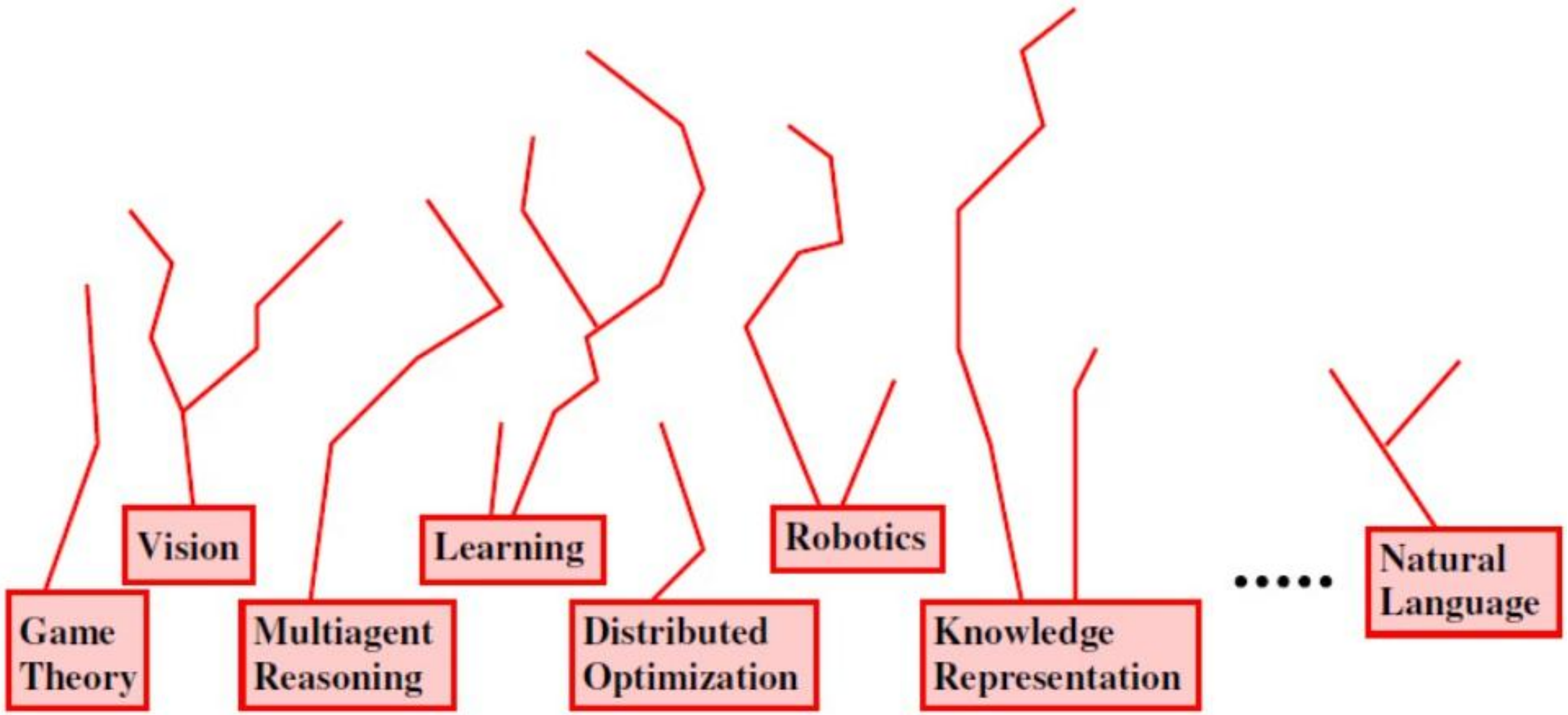
Bottom-up metaphor:

- *Russell, '95: "Theoreticians can produce the AI equivalent of bricks, beams, and mortar with which AI architects can build the equivalent of cathedrals.*

Bottom-Up Approach

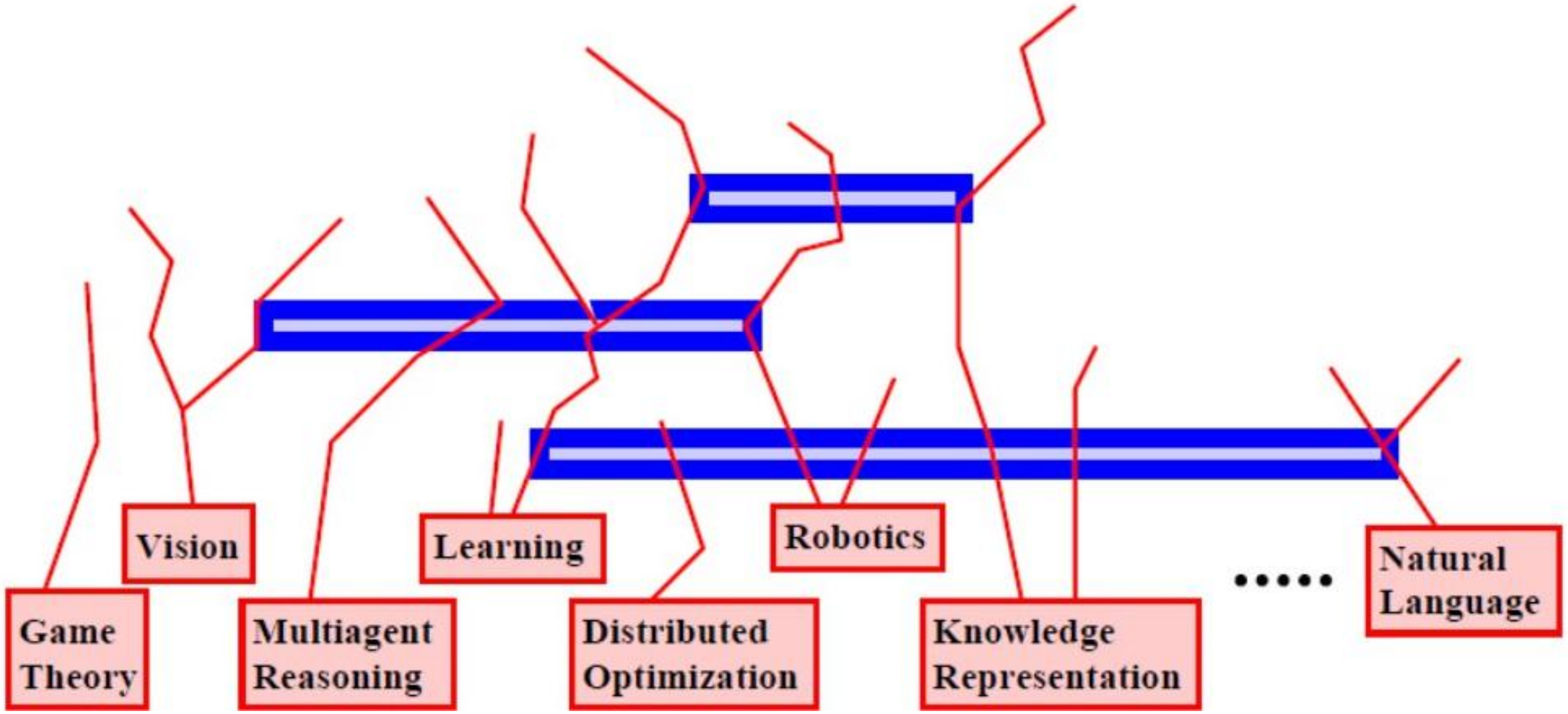


Bottom-Up Approach – The Bricks



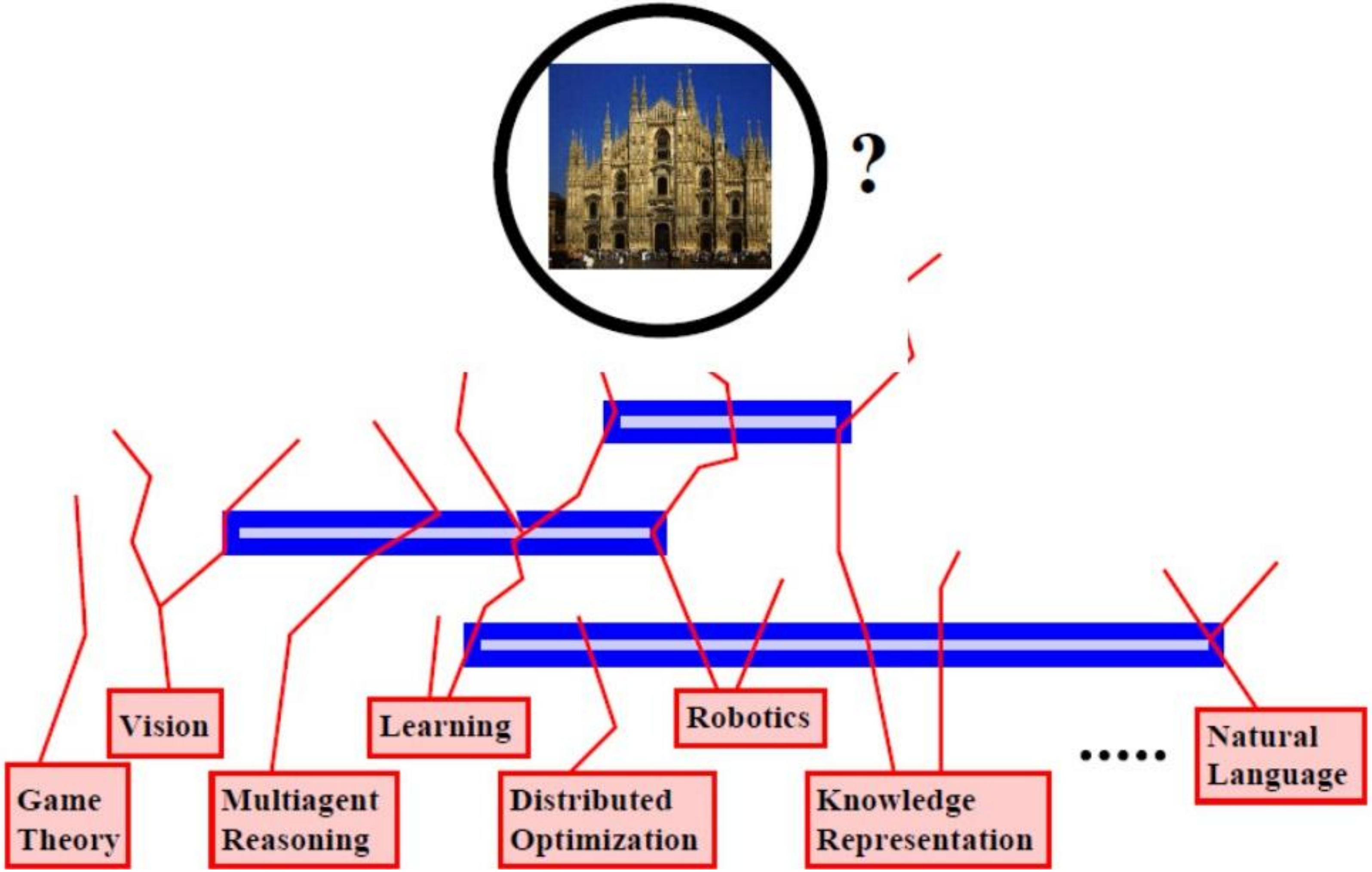
- Course Format
- Introduction to AI

Bottom-Up Approach – The Beams and Mortar

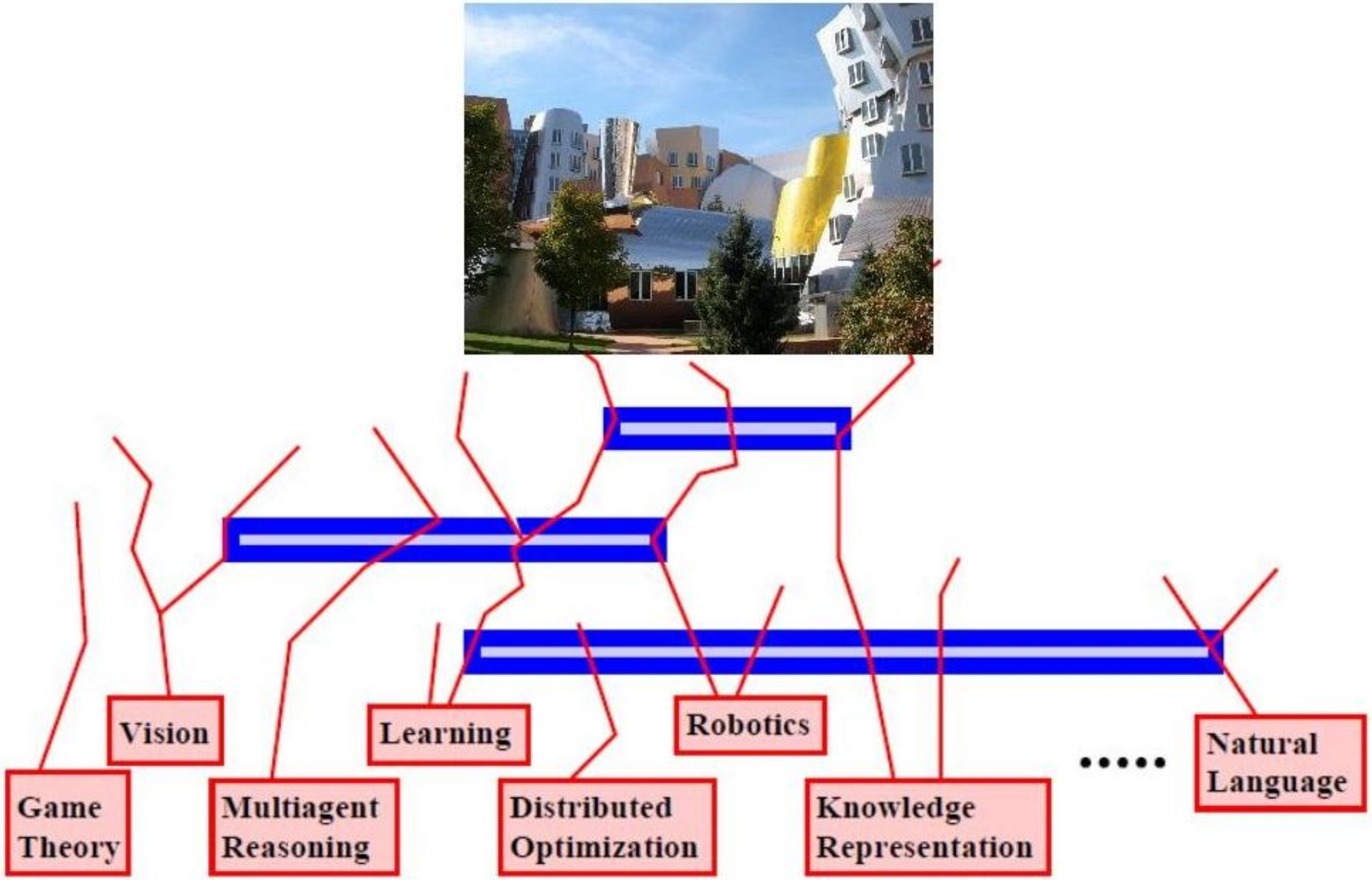


- Course Format
- Introduction to AI

Bottom-Up Approach – Towards a Cathedral?



Bottom-Up Approach – Or Something Else?



- Course Format
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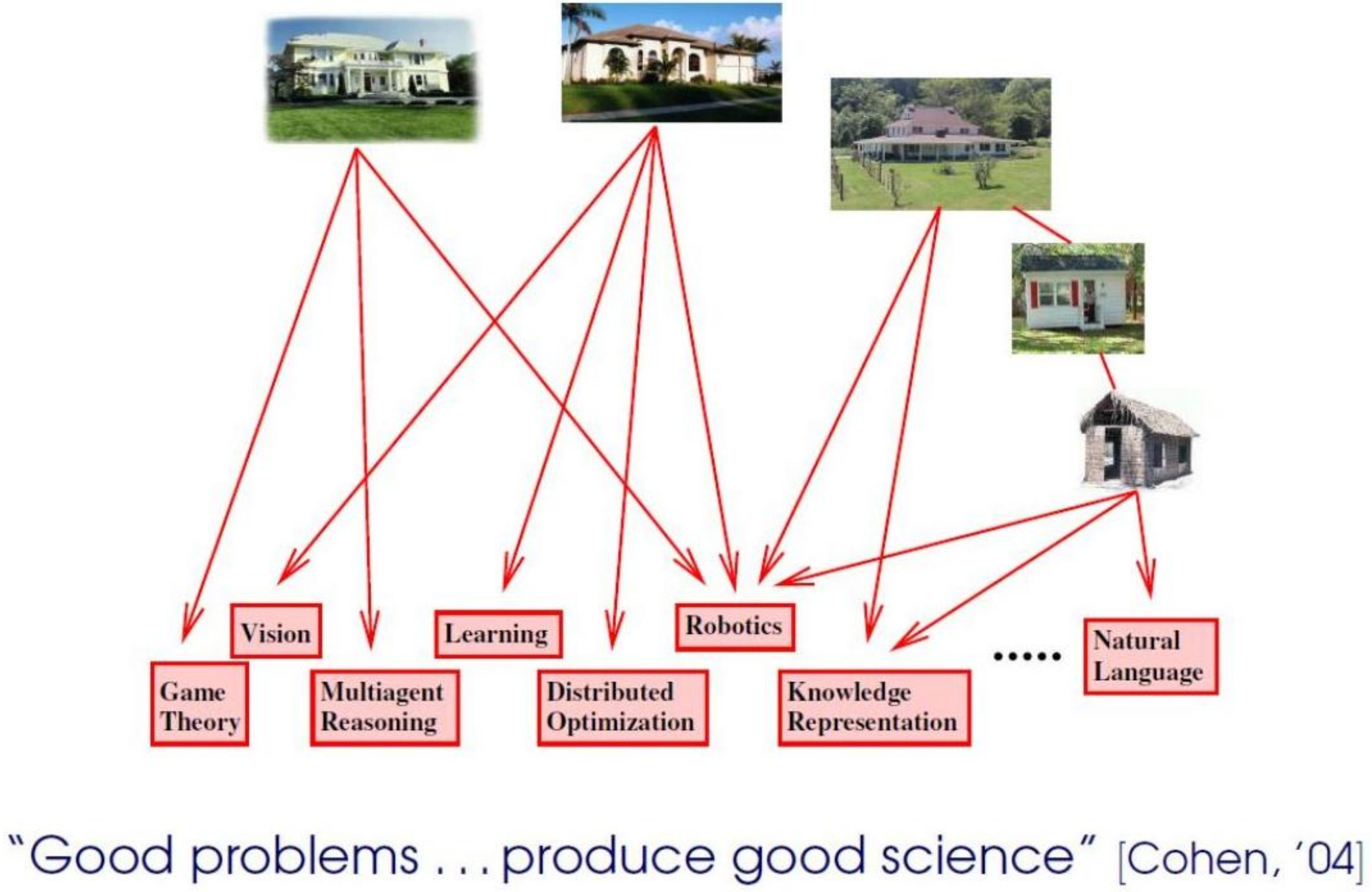
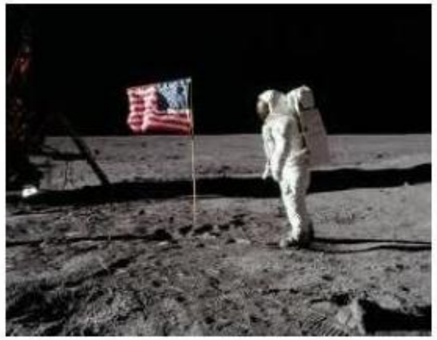
Top-Down Approach



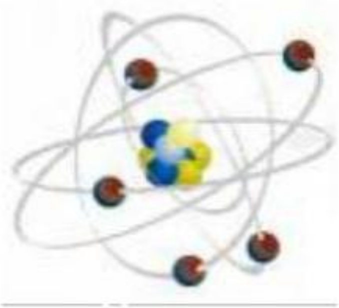
Introduction

Top-Down Approach

Manned flight



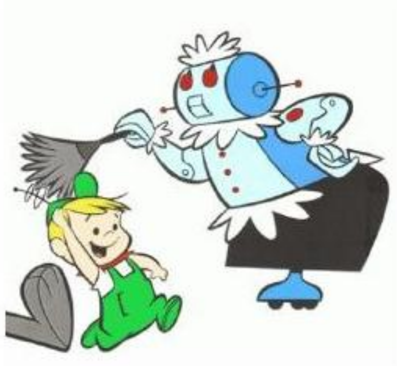
Apollo mission



Manhattan project

Autonomous vehicles

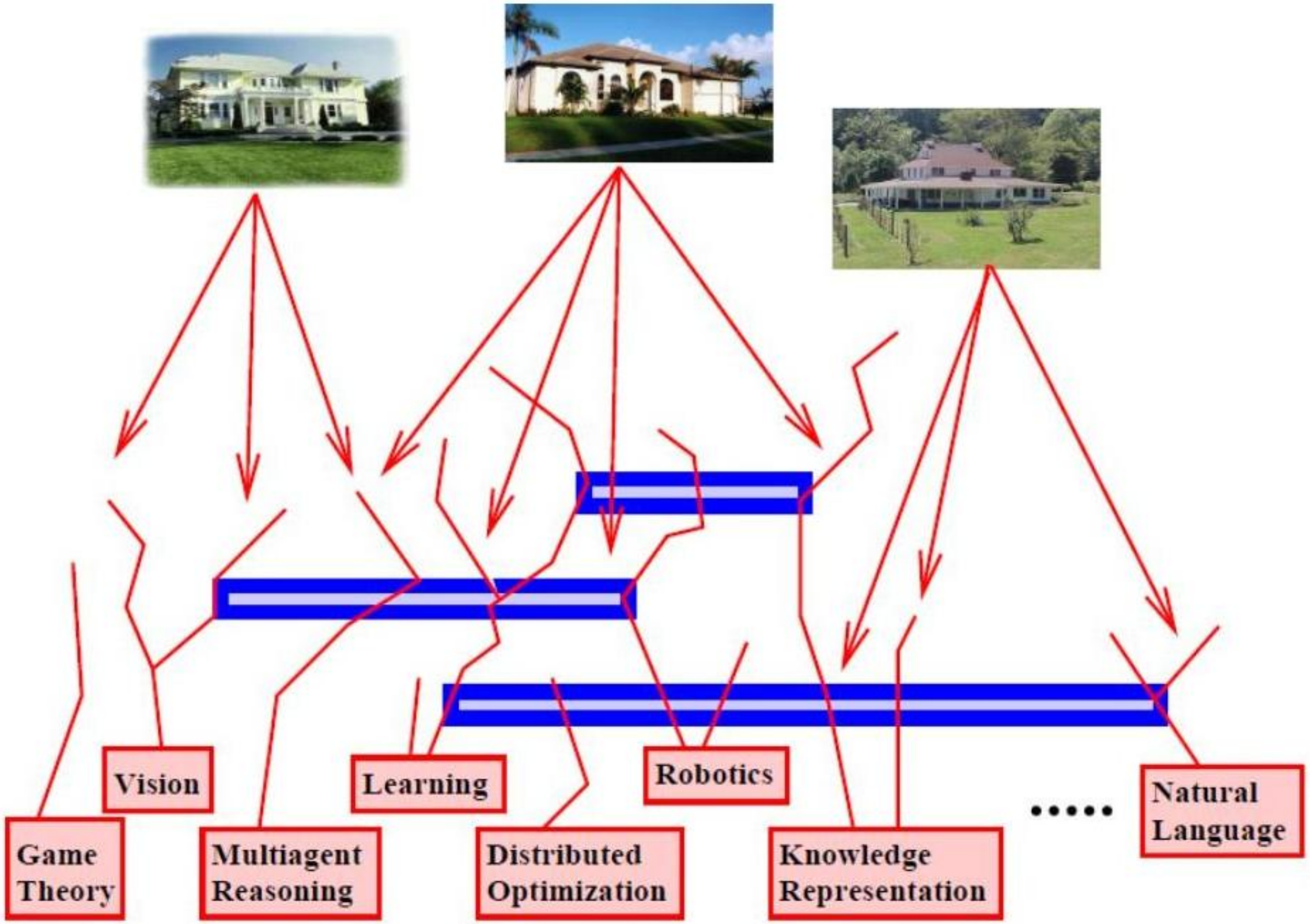
RoboCup soccer



Assistive robots

"Good problems . . . produce good science" [Cohen, '04]

Meeting in the Middle



Implications of AI

- **A goal of AI:** Robust, fully autonomous agents in the real world

What happens when we achieve this goal?

?

?

Is AI moving us in the right direction?

Discussion Question: Would you rather have been born:

– 50 years earlier? – 50 years later?

Introduction

Difficult Questions?

- Who is liable if a robot driver has an accident?
- If machines surpass human intelligence (in all ways)?
- Could such machines have conscious existence? Rights?
- What IS a mind?
- How can a physical Object have a mind?
- Can we build a mind?

AI is one of the great intellectual adventures of 20th and 21st centuries!

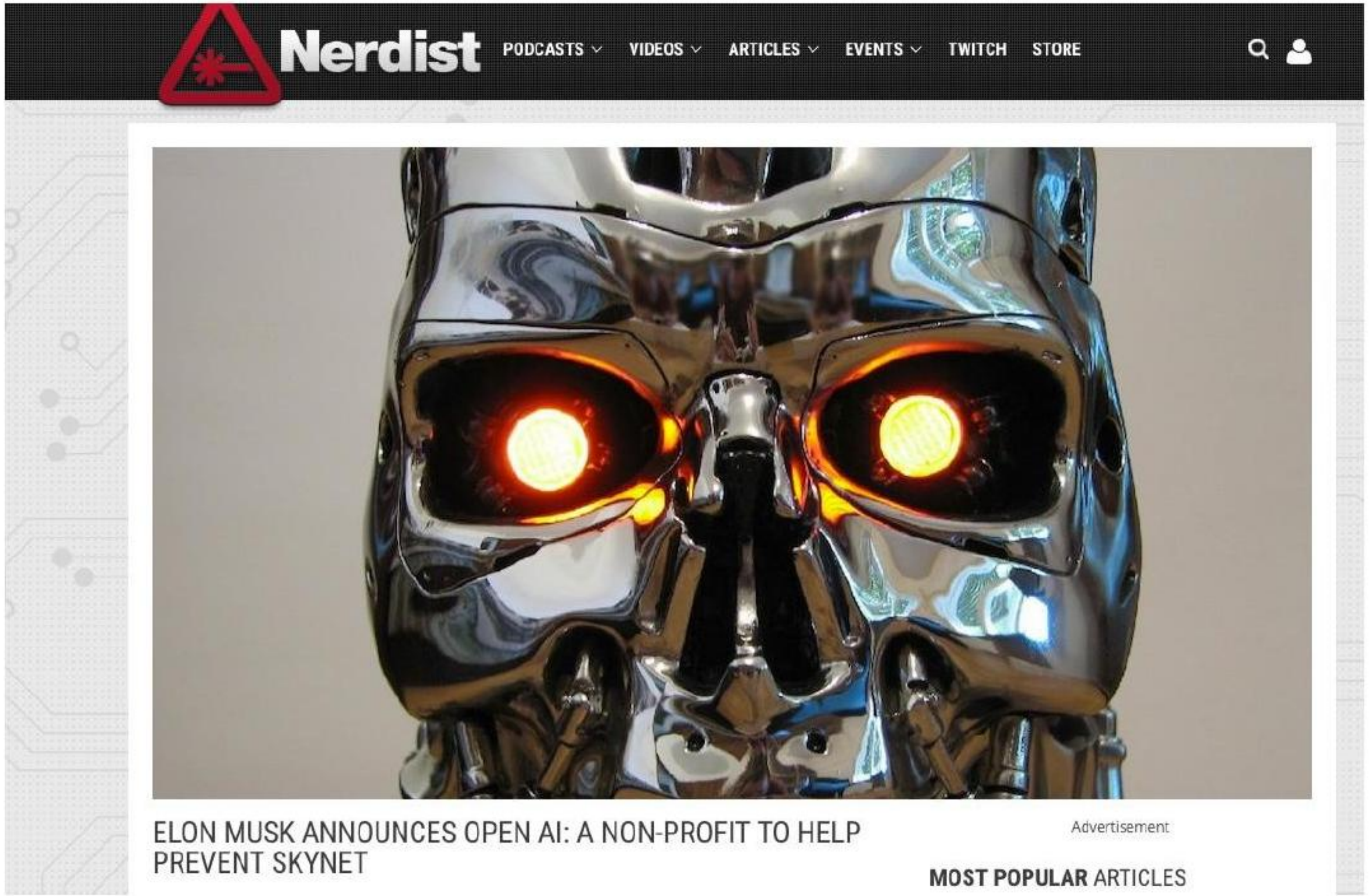
What is AI?

- Finding fast algorithms for NP-hard problems
- Getting computers to do the things they can’t do yet
- Getting computers to do the things they do in the movies

Sci-Fi AI

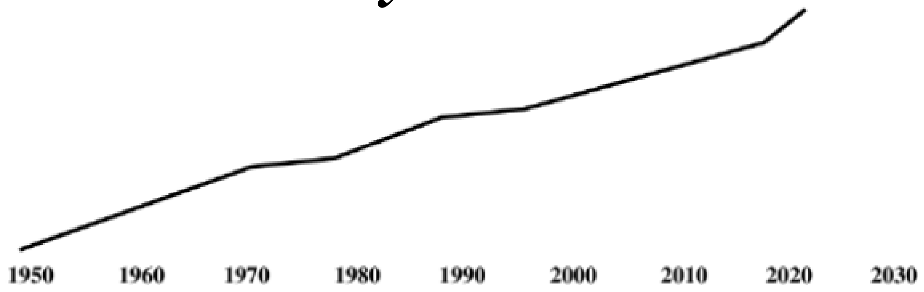
- Course Format
- Introduction to AI

AI in the News

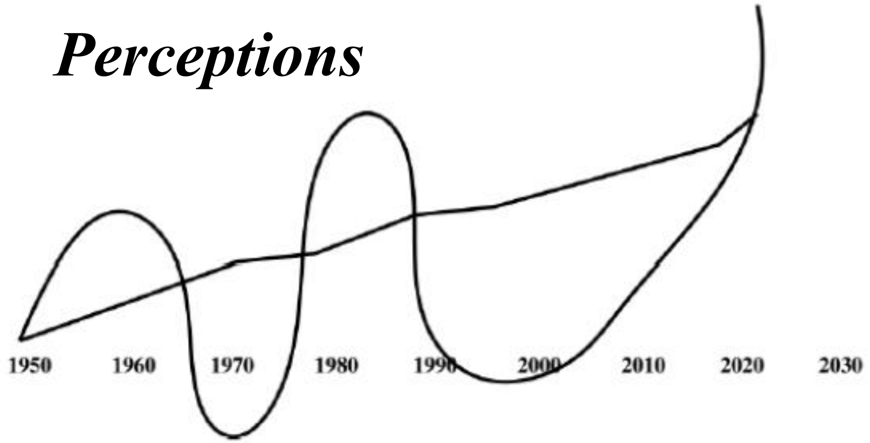


AI - Hype

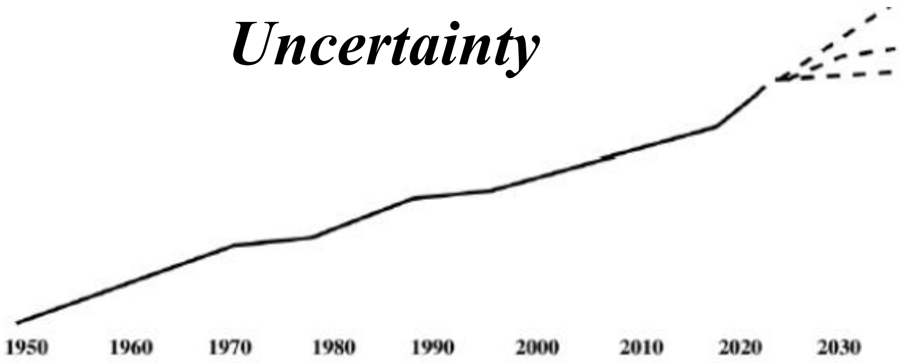
Reality



Perceptions



Uncertainty



Perception Uncertainty